

Free Groups

Vincent Wang

November 2025

1 Free Groups

Let G be a group and $\{G_\alpha\}$ be a collection of subgroups. A function $w : [n] \rightarrow G$ for some $n \in \mathbb{Z}^+$ and $w(i) \in G_\alpha$ for some $\alpha \in J$, for all $i \in [n]$ is called a word in $\{G_\alpha\}$. Note that the empty function is a word. Denote by W the set of all words in $\{G_\alpha\}$.

There exists an intrinsic function $l : W \rightarrow \mathbb{Z}_{\geq 0}$ given by $l(w) = |\text{dom } w|$. Let $w, w' \in W$ and $n = l(w), n' = l(w')$. Define the append function $A : W \times W \rightarrow W$ to be the word $A(w, w') : [n + n'] \rightarrow G$ given by:

$$A(w, w')(i) = \begin{cases} w(i) & \text{for } i \in [n], \\ w'(i - n) & \text{for } i \in [n + n'] - [n]. \end{cases}$$

One can check that A is well-defined and $l(A(w, w')) = l(w) + l(w')$ (by induction).

Define the evaluation function $E : W \rightarrow G$ as follows: Suppose $n = l(w)$. Let $e_0 = 1$ and $e_{i+1} = e_i \cdot w(i + 1)$ for $i \in [n - 1]$. Take $E(w) = e_n$.

I claim that $E(A(w, w')) = E(w)E(w')$. We proceed by induction on n' . If $n' = 0$, we're done. $E(A(w, w') \frown x) = E(A(w, w')) \cdot x = E(w)E(w')x = E(w)E(w' \frown x)$.

Let $G = G_1 * G_2$ and $N_1 \triangleleft G_1, N_2 \triangleleft G_2$ normal subgroups. Denote by N the least normal subgroup containing G_1, G_2 . Then $G/N \cong (G_1/N_1) * (G_2/N_2)$.

Proof: For notation, denote by $H = (G_1/N_1) * (G_2/N_2)$. Consider the composition of maps $h_1 : G_1 \rightarrow G_1/N_1 \rightarrow H$. Similarly consider the composition $h_2 : G_2 \rightarrow H$. By extension property of G , we can extend to $G \rightarrow H$. Since the projection map $G_1 \rightarrow G_1/N_1$ is surjective, G_1/N_1 and G_2/N_2 generate H , the map $G/N \rightarrow H$ is surjective.

Observe that any element of N_1 or N_2 is mapped by either h_1 or h_2 to the identity element of $(G_1/N_1) * (G_2/N_2)$, so $\ker \phi$ contains N_1, N_2 . Then $\ker \phi =$

$\mathcal{N}(\ker \phi) > \mathcal{N}(N_1, N_2) = N$, so we may descend to a surjective map $\phi : G/N \rightarrow (G_1/N_1) * (G_2/N_2)$.

Conversely, consider the composition of maps $G_1 \rightarrow G \rightarrow G/N$. Since N_1 is mapped to the identity, we descend to a map $G_1/N_1 \rightarrow G/N$. Similarly, we have the map $G_2/N_2 \rightarrow G/N$, so by extension property we extend to $\psi : (G_1/N_1) * (G_2/N_2) \rightarrow G/N$. It is enough to show that ψ is the left inverse of ϕ .

Since G_1, G_2 generate G , it is enough to show that $\psi\phi = \text{id}_{G/N}$ over G_1, G_2 . We have for $g_1 \in G_1$, $\psi\phi(g_1N) = \psi(g_1N_1) = g_1N$. Similarly, $\psi\phi(g_2N) = g_2N$ for $g_2 \in G_2$, AWD.

Munkres's Proof: Consider the composition of maps $G_1 \rightarrow G \rightarrow G/N$. Since N_1 is mapped to the identity, we descend to $i_1 : G_1/N_1 \rightarrow G/N$. Similarly, we have the composition $i_2 : G_2/N_2 \rightarrow G/N$.

Suppose $h_1 : G_1/N_1 \rightarrow H, h_2 : G_2/N_2 \rightarrow H$ are homomorphisms. I show that $G/N \cong (G_1/N_1) * (G_2/N_2)$ by showing the extension property: there exists a homomorphism $h : G/N \rightarrow (G_1/N_1) * (G_2/N_2)$ such that $hi_1 = h_1, hi_2 = h_2$.

Let $k_1 = h_1p_1 : G_1 \rightarrow H$. Define $k_2 : G_2 \rightarrow H$ similarly. By extension property of G , let $k : G \rightarrow H$ extend k_1, k_2 . Note that N_1, N_2 are mapped by k to the identity of H , so k descends to a map $h : G/N \rightarrow H$.

2 Properties of Free Groups

Given a collection of groups $\{G_\alpha\}$, G is said to be the external free product of the groups $\{G_\alpha\}$ if there exist homomorphisms $\{i_\alpha : G_\alpha \rightarrow G\}$ such that the groups $i_\alpha(G_\alpha)$ generate G and $\emptyset$ is the only reduced word that evaluates to 1.

Claim: each i_α is a monomorphism. Suppose otherwise, so let $x_\alpha \in \ker i_\alpha$ not equal the identity. Then the reduced word (x_α) evaluates to 1.

Claim: the $\{i_\alpha(G_\alpha)\}$ need to generate G . Otherwise, take a single trivial group 1 and include $1 \hookrightarrow G$. The only reduced word in 1 is $\emptyset$, but we can take $G \neq 1$.

Claim: let $\{G_\alpha\}$ be a collection of groups, $\{i_\alpha : G_\alpha \rightarrow G\}$ a collection of monomorphisms into a free product G . Suppose $\{i'_\alpha : G_\alpha \rightarrow G'\}$ is a another collection of monomorphisms into another free product G' . Then $G \cong G'$.